

Opportunity Dataset

Exploratory Data Analysis Report

Prepared by: Team 1

Program: Excelerate; Data Visualization Trainee Program

Week 2 Deliverable | 0308 SLU DVT Team 1

Source: Team1_Cleaned_Opportunity_Dataset.xlsx (Week 1 output)

1. Exploratory Data Analysis Report

1.1 Overview of the Dataset

This report builds directly on the Week 1 cleaned dataset (Team1_Cleaned_Opportunity_Dataset.xlsx), which retained all 5,728 non-corrupted records from the original export and flagged 4,186 of them as suspected test, demo, or automation-generated entries using the `is_test_or_placeholder` column.

For this analysis, the team excluded flagged records and worked from the remaining 1,542 production-representative opportunities. This is a deliberate methodology choice, not a re-cleaning of the data: the Week 1 file keeps every row for auditability, and Week 2 applies a filter on top of it so that trends, locations, and scholarship figures reflect genuine opportunities rather than internal test noise.

Metric	Value
Records in Week 1 cleaned file	5,728
Flagged as test/placeholder (excluded here)	4,186
Records analysed in this report	1,542
Date range covered	August 2022 – May 2026
Categories represented	10 (Internship, Career, Competition, Course, Event, Engagement, Masterclass, JobSimulation, Program, Xploreu)

1.2 Data Availability Note

Before presenting findings, it's important to flag a gap between the Week 2 brief and the fields actually present in the Opportunity dataset. The brief asks for analysis of "applications by location" and "outreach channel performance." The dataset contains one row per opportunity, not per application or per applicant, and it does not include any channel/source field indicating how participants discovered an opportunity.

- Applications by location: analyzed as opportunity volume by location instead; i.e. how many opportunities are offered per location, not how many people applied to them. This is the closest available proxy and is labelled as such throughout Section 1.4.
- Outreach channel performance: could not be analyzed. No column in the dataset records a discovery or acquisition channel. This is called out as a data gap and recommendation in Section 3.5 rather than estimated.

1.3 Analysis of Opportunity Sign-Up Trends

Opportunity volume is not steady over time. There is a sharp jump in early 2023 (228 opportunities in Q1 2023 alone, up from 14–2 in the prior two quarters), a gradual decline through late 2024, and a second, larger spike in Q4 2025 (263

opportunities; the single highest month is October 2025) before tapering into 2026. Year-over-year, 2024 fell 38.4% from 2023's total of 557, before 2025 recovered with a 27.7% rise back to 438. The Q4 2024 low (34 opportunities) is the quietest quarter in the dataset outside of the initial 2022 ramp-up. This pattern; sharp spikes followed by quiet periods rather than steady month-on-month growth; suggests opportunities are added in batches, most likely tied to specific recruitment cycles, partner onboarding, or seasonal program launches, rather than a continuous publishing pipeline. See Figure 1 in Section 2 for the full quarterly trend.

1.4 Analysis of Applications by Location

As noted in Section 1.2, this analysis reflects opportunity volume per location, not individual application counts, since the dataset is opportunity-level. Work From Home (644) and Virtual (509) together account for the overwhelming majority of opportunities with a defined location; remote-format opportunities outnumber every other location combined. A substantial 387 records (25% of the analyzed set) have no location specified at all, which is worth flagging as a data completeness gap rather than a true "in-person" category. Only two records point to a specific physical location (Kolkata) or contain leftover placeholder text ("Quis voluptate totam") that the Week 1 cleaning pass did not catch; a minor residual data quality issue worth a follow-up correction. See Figure 2 in Section 2.

1.5 Analysis of Outreach Channel Performance

This analysis could not be performed. The table below sets out exactly what would be required and what the dataset currently provides.

Required Analysis	Supported by Dataset?	What Would Be Needed
Outreach channel identification	No	A channel/source field on each opportunity record
Applications by channel	No	Application-level records, not just opportunity-level records
Channel conversion/performance	No	An application or conversion outcome field per channel

This is treated as a confirmed data gap rather than an assumed value; see the recommendation in Section 3.5 to request this field be added to future exports.

1.6 Analysis of Scholarship Distribution

The microscholarship field records a scholarship value associated with 1,541 of the 1,542 analyzed opportunities. Two records carry extreme values (500,000,000 and 10,000,000) that are inconsistent with every other value in the field (the next highest is 3,513) and follow the same outlier pattern already documented for the fee field in the Week 1 cleaning log. These two records were excluded from the analysis below and flagged for the same manual review recommended for fee outliers; the median, not the mean, is used throughout this section since it is not distorted by them. Course (median 120) and Competition/Masterclass (median 100) carry the highest typical scholarship values, while JobSimulation carries the lowest (median 6). Internship, the largest category by volume, sits in the lower-middle of the scholarship range (median 42); the platform's highest-volume opportunity type is not where the largest scholarship value is concentrated. See Figure 5 and the summary table in Section 2.

1.7 Scholarship Availability

Beyond the value of the scholarship itself, it's worth checking how many opportunities carry a scholarship at all. Of the 1,541 opportunities with a recorded scholarship status, 1,148 (74.5%) have a positive scholarship value and 393 (25.5%) are recorded as zero. Availability is not evenly spread across categories: within the six largest categories, Internship has the widest gap in absolute terms (591 with a scholarship vs. 212 without), while the smaller categories skew slightly more toward having a scholarship, proportionally, than Internship does. See Figure 6 and Figure 7 in Section 2.

1.8 Relevant Summary Statistics

A category-level summary is provided as Table 1 in Section 2, combining opportunity volume, median scholarship value, and average paid fee in a single reference table.

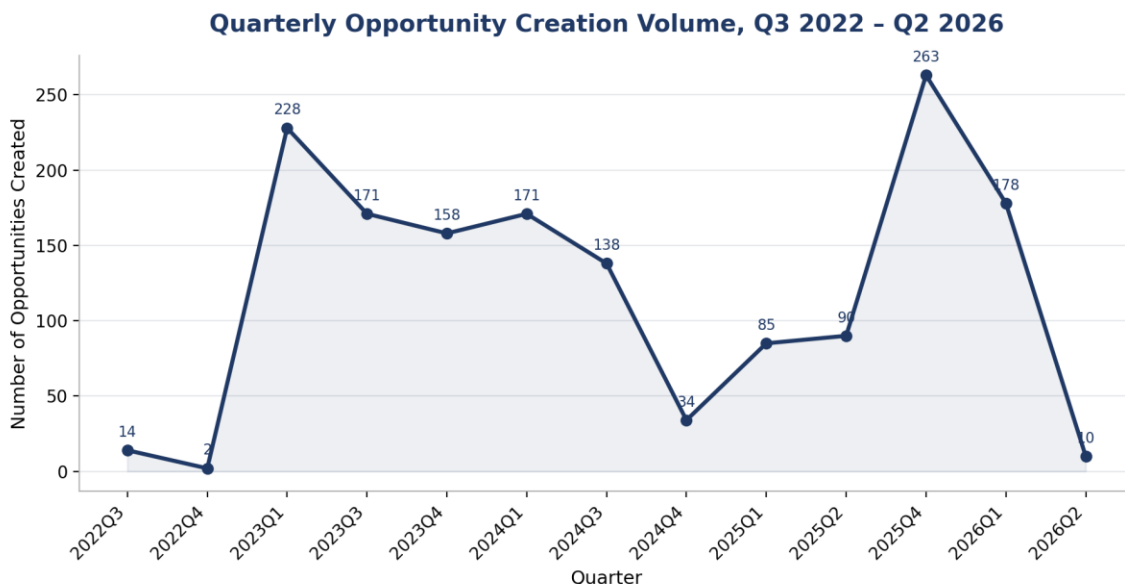
1.9 Relationship Analysis: Category and Location

Beyond looking at category and location separately, it's worth checking whether the two interact; for example, whether certain categories skew more toward Virtual or Work From Home delivery. Restricting to the 1,153 records with one of the two standardized location values (509 Virtual + 644 Work From Home), Work From Home is consistently more common than Virtual within every major category, not just overall. Internship shows the largest gap in absolute terms (322 Work From Home vs. 266 Virtual), while Masterclass shows the largest relative gap (33 Work From Home vs. 11 Virtual; exactly three to one). This suggests the Work From Home/Virtual split is a broadly consistent platform-wide pattern rather than something driven by any one category. See Figure 8 and Table 2 in Section 2.

2. Visual Analysis Section

This section presents every chart and table supporting the findings above. Charts use a single consistent navy palette throughout; including a lighter navy tone for two-series comparisons, matching the same blues used in the pie chart; are sorted from highest to lowest where categories are being ranked, and are labeled with the metric, unit, and sample size (n) on each axis. Each visualization is followed by a short explanation that leads with the finding rather than restating the chart.

Figure 1; Quarterly Opportunity Creation Volume, Q3 2022 – Q2 2026

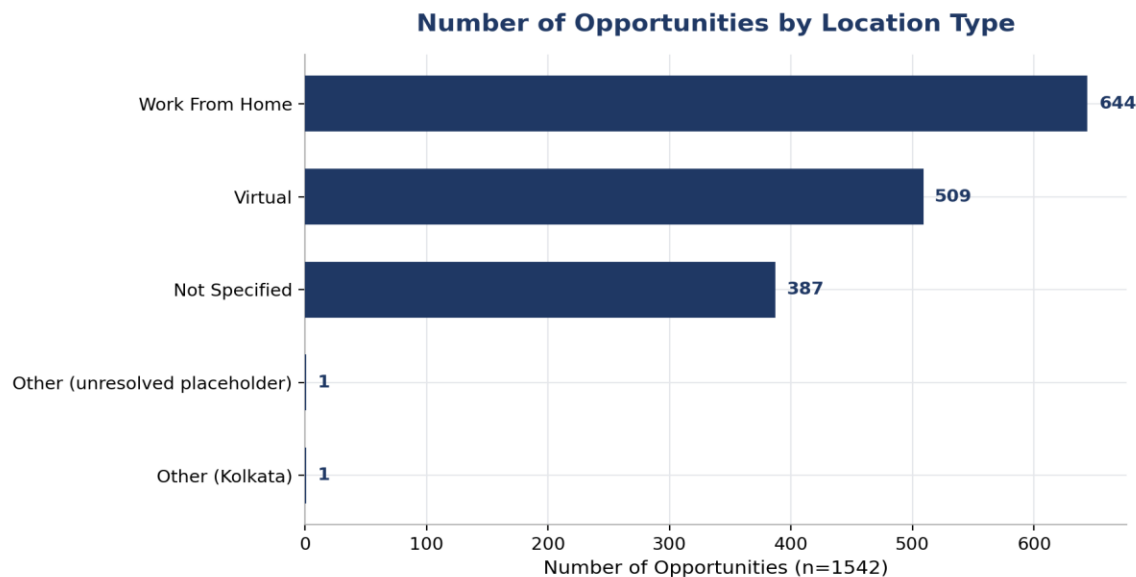


Line chart. Y-axis: number of opportunities created per quarter. X-axis: calendar quarter.

Creation activity moves in two sharp cycles rather than growing steadily: a peak of 228 in Q1 2023 and a larger peak of 263 in Q4 2025, separated by a sustained decline through 2024 that bottoms out at just 34 in Q4 2024. The size and timing of these two peaks; roughly two years apart; is the strongest signal in the dataset that opportunities are published in scheduled batches rather than continuously.

Quarter	Opportunities Created	Share of Dataset
2022 Q3	14	0.9%
2022 Q4	2	0.1%
2023 Q1	228	14.8%
2023 Q3	171	11.1%
2023 Q4	158	10.2%
2024 Q1	171	11.1%
2024 Q3	138	9.0%
2024 Q4	34	2.2%
2025 Q1	85	5.5%
2025 Q2	90	5.8%
2025 Q4	263	17.1%
2026 Q1	178	11.5%
2026 Q2	10	0.6%

Figure 2; Number of Opportunities by Location Type

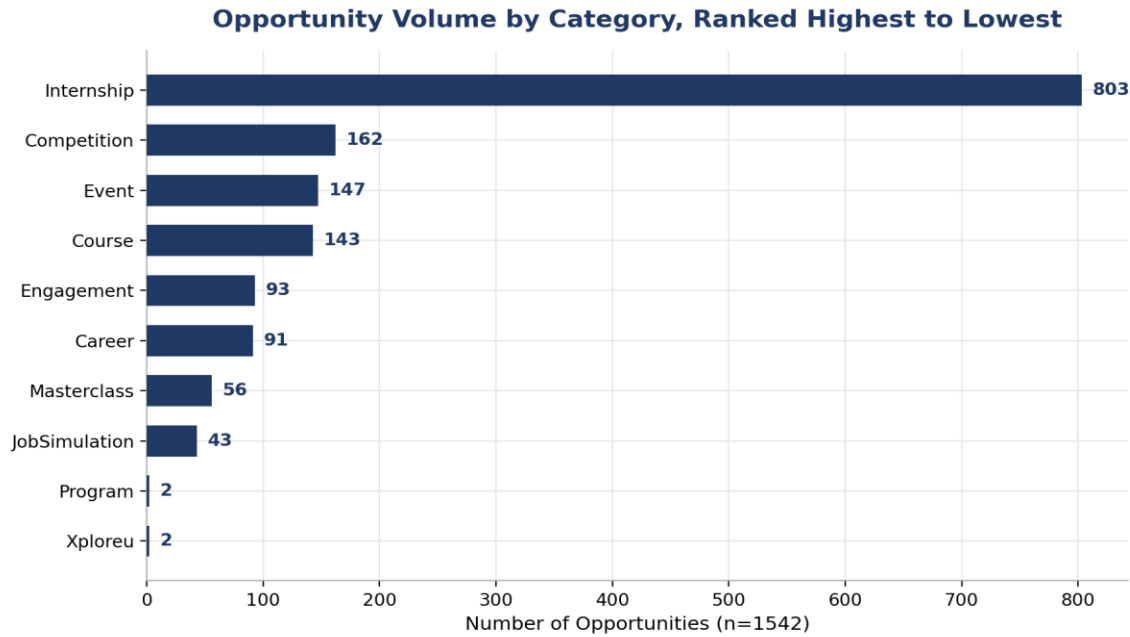


Horizontal bar chart, ranked highest to lowest. X-axis: number of opportunities (n=1,542).

Work From Home (644) edges out Virtual (509) as the single largest location type, and together the two remote formats account for 75% of all opportunities with a recorded location. The 387 "Not Specified" records (25% of the full set) are large enough on their own to outrank every category except the top two; this is a data completeness issue worth fixing before location is used for any downstream reporting.

Location	Number of Opportunities	Share of Dataset
Work From Home	644	41.8%
Virtual	509	33.0%
Not Specified	387	25.1%
Other (Kolkata)	1	0.1%
Other (unresolved placeholder)	1	0.1%

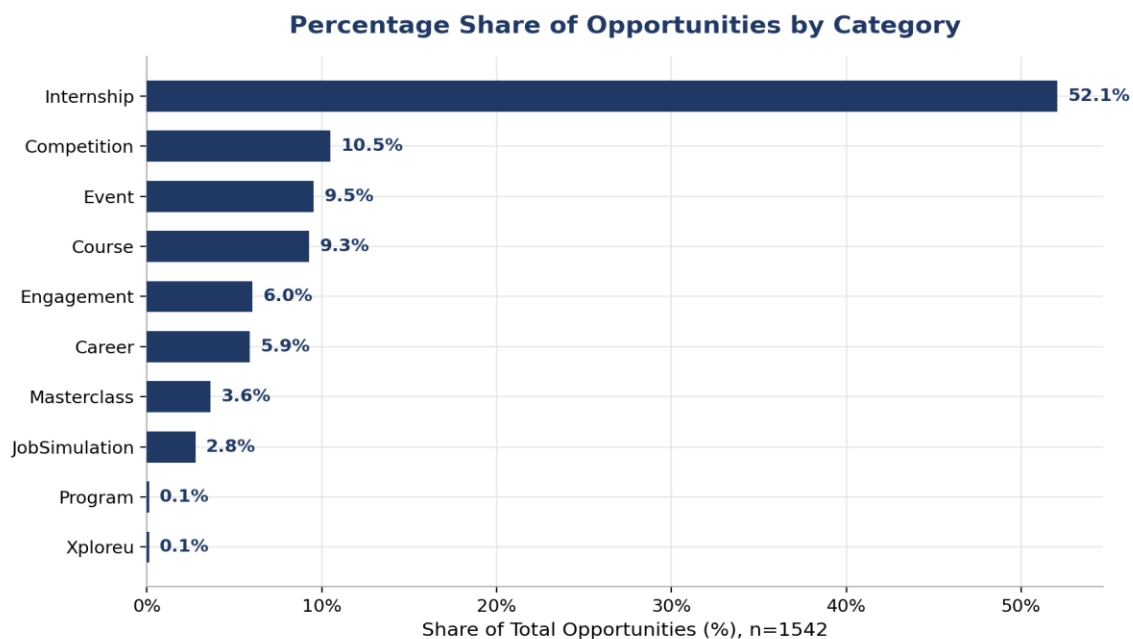
Figure 3; Opportunity Volume by Category, Ranked Highest to Lowest



Horizontal bar chart, sorted descending. X-axis: number of opportunities (n=1,542).

Internship (803) is more than five times larger than the next category, Competition (162); a concentration this steep means platform-wide metrics will move almost entirely in line with whatever is happening inside the Internship category specifically, regardless of trends in the other nine categories.

Figure 4; Percentage Share of Opportunities by Category

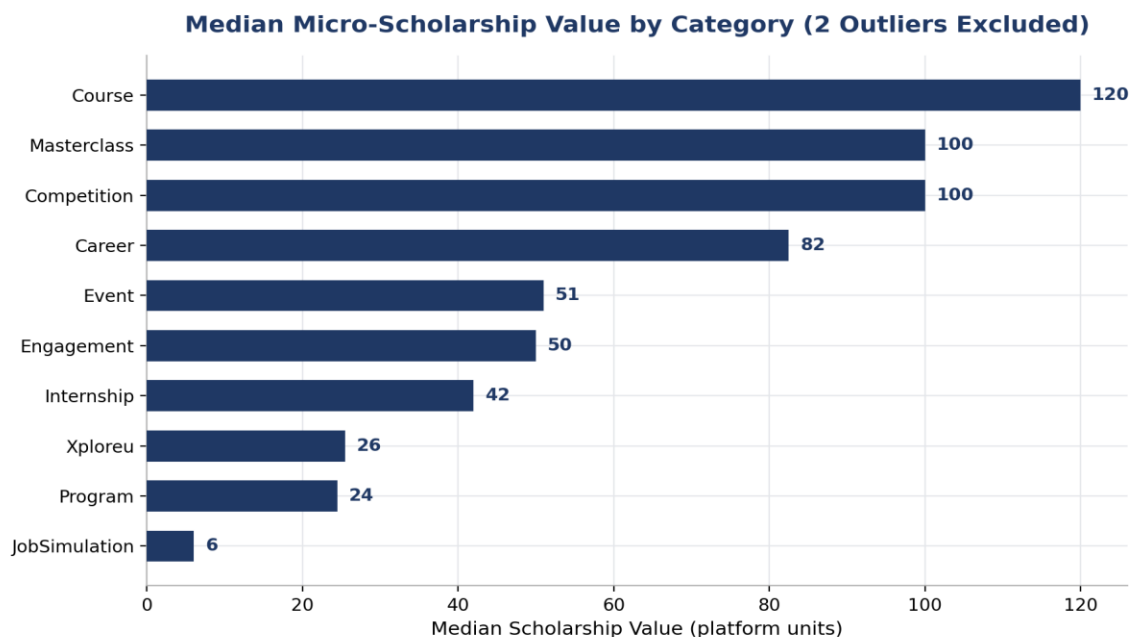


Horizontal bar chart, sorted descending. X-axis: share of total opportunities (%), n=1,542.

Internship alone represents 52.1% of all analyzed opportunities; a majority share on its own. The bottom five categories (Masterclass, JobSimulation, Program, and Xploreu) combined make up less than 7% of the dataset, confirming this is a heavily top-weighted distribution, not a roughly even spread across ten categories.

Category	Number of Opportunities	Share of Dataset
Internship	803	52.1%
Competition	162	10.5%
Event	147	9.5%
Course	143	9.3%
Engagement	93	6.0%
Career	91	5.9%
Masterclass	56	3.6%
JobSimulation	43	2.8%
Program	2	0.1%
Xploreu	2	0.1%

Figure 5; Median Micro-Scholarship Value by Category (2 Outliers Excluded)



Horizontal bar chart, sorted descending. X-axis: median scholarship value in platform units.

Course carries the highest typical scholarship value (120), roughly 20 times higher than JobSimulation (6), the lowest. Internship; the category with by far the most volume; sits below the midpoint of this ranking at 42, meaning scholarship value is not allocated in proportion to how many opportunities a category has.

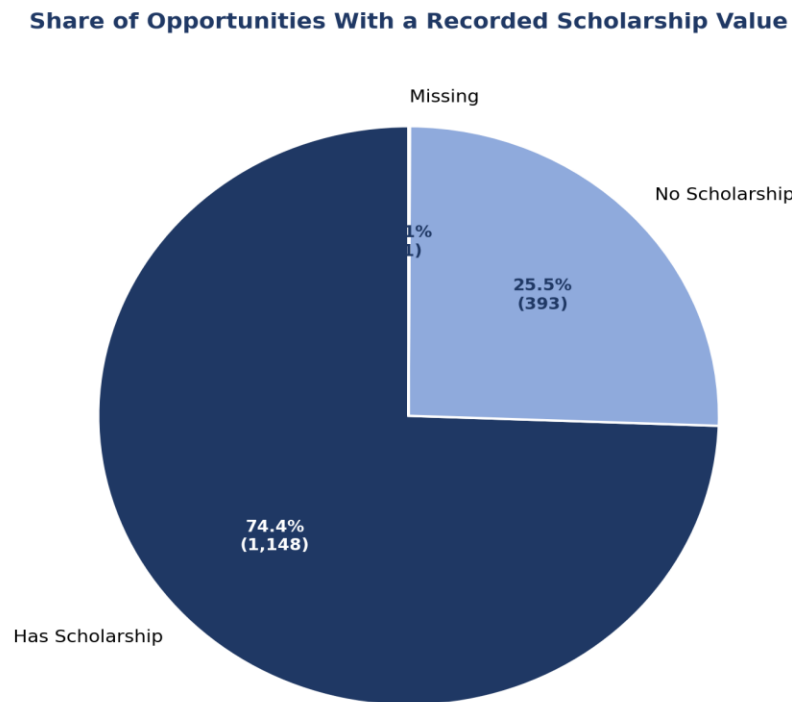
Category	Median Scholarship Value
Course	120.0
Competition	100.0
Masterclass	100.0
Career	82.5
Event	51.0
Engagement	50.0
Internship	42.0
Xploreu	25.5

Program	24.5
JobSimulation	6.0

Table 1; Summary Statistics by Category

Category	Opportunities	Median Scholarship	Avg. Paid Fee (where fee > 0)
Internship	803	42.0	354
Competition	162	100.0	42
Event	147	51.0	53
Course	143	120.0	24
Engagement	93	50.0	32
Career	91	82.5	11,143*
Masterclass	56	100.0	33
JobSimulation	43	6.0	45
Program	2	24.5	52
Xploreu	2	25.5	38

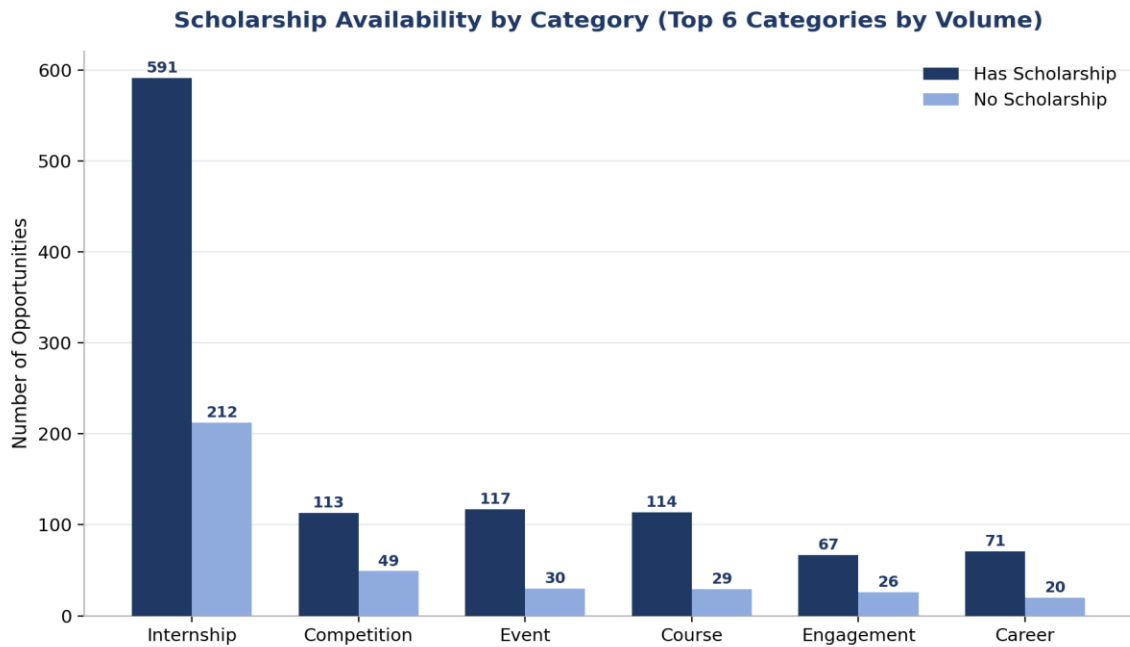
*Career's average paid fee is heavily skewed by a single 200,000 USD record already flagged for review in the Week 1 fee_flag column; the underlying median paid fee for Career is far lower.

Figure 6; Share of Opportunities With a Recorded Scholarship Value

Pie chart. Two categories only: Has Scholarship vs. No Scholarship (n=1,541; 1 record with missing scholarship status excluded from this chart).

Three in four opportunities (74.5%) carry a positive scholarship value, meaning "no scholarship" is the minority case, not the norm, across the platform as a whole.

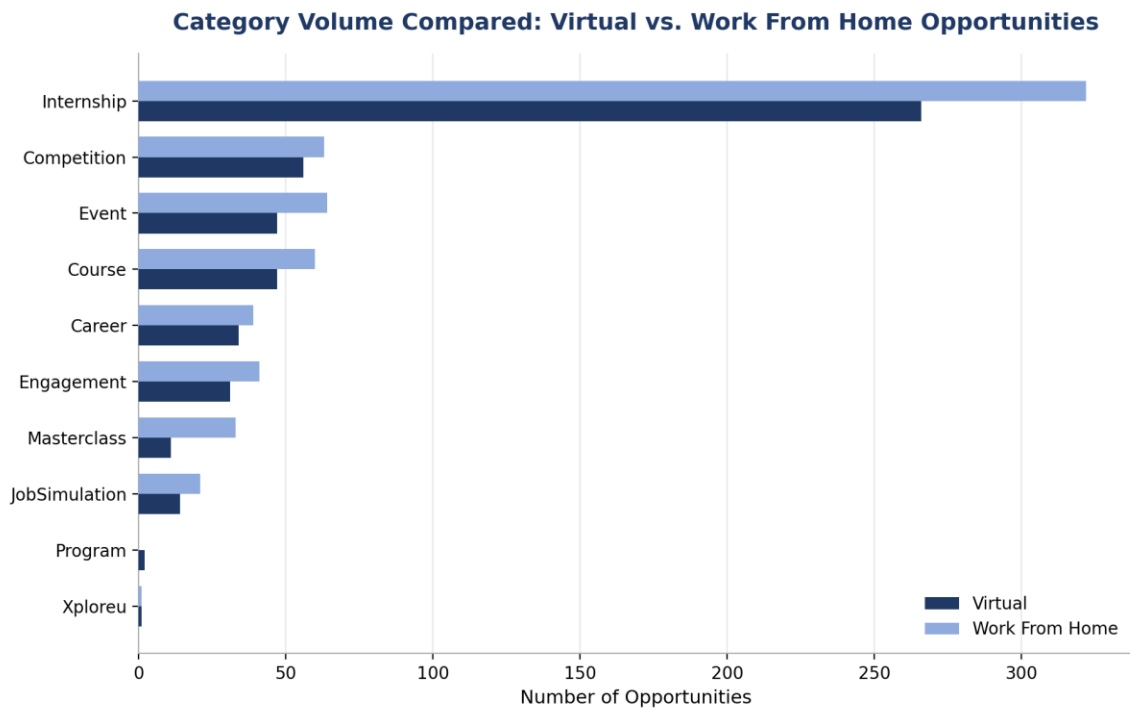
Figure 7; Scholarship Availability by Category (Top 6 Categories by Volume)



Vertical clustered column chart. Y-axis: number of opportunities. Categories limited to the top 6 by volume for readability.

Internship has by far the largest volume without a scholarship in absolute terms (212), but that's expected given its overall size; proportionally, roughly 26% of Internship opportunities lack a scholarship, close to the 25.5% platform-wide rate shown in Figure 6. No category in the top 6 stands out as disproportionately under-scholarshipped once volume is accounted for.

Figure 8; Category Volume Compared: Virtual vs. Work From Home Opportunities



Horizontal grouped bar chart, sorted by total volume. X-axis: number of opportunities.

Work From Home outnumbers Virtual in every category shown, not just overall; the gap is widest in Internship (322 vs. 266, a difference of 56) and proportionally widest in Masterclass (33 vs. 11, a 3-to-1 ratio). This consistency across categories of very different sizes is what makes the Work From Home preference a platform-wide pattern rather than something specific to any one category.

Table 2; Category by Location Cross-Tabulation

Category	Virtual	Work From Home
Internship	266	322
Competition	56	63
Event	47	64
Course	47	60
Career	34	39
Engagement	31	41
Masterclass	11	33
JobSimulation	14	21
Program	2	0
Xploreu	1	1

3. Key Findings Summary

3.1 Major Trends Identified

- Opportunity volume is cyclical, not steady; large batch additions in Q1 2023 and Q4 2025, separated by an extended lull through 2024.
- Remote-format opportunities (Work From Home + Virtual) dominate, accounting for 75% of all opportunities with a specified location.

3.2 Important Patterns or Relationships

- Internship is the platform's core offering by volume (52% of production records) but carries only a mid-range scholarship value, while lower-volume categories like Course and Competition offer proportionally higher median scholarships.
- The microscholarship field shows the same extreme-outlier pattern already identified in the fee field during Week 1 cleaning; a signal this may be a systemic data entry issue rather than isolated incidents.
- Work From Home outnumbers Virtual within every major category, not just overall; the remote-format skew is a consistent platform-wide pattern rather than being driven by any single category.

3.3 Noteworthy Observations

- A quarter of records have no location specified; a completeness gap that limits how confidently location-based conclusions can be drawn.
- Career's average paid fee is distorted by a single already-flagged outlier record, underscoring why the Week 1 fee_flag column matters for downstream analysis.

3.4 Potential Areas for Deeper Investigation

- Outreach channel performance could not be assessed; this field does not exist in the current dataset and represents a genuine data gap.

- Investigate the cause of the Q1 2023 and Q4 2025 volume spikes; if repeatable, they could inform a publishing calendar; if one-off imports, they should be excluded from any "average opportunities per quarter" metric.
- Manually verify the two extreme microscholarship values before they are used in any dashboard or reporting.
- Examine whether the Career category's fee distribution reflects genuine premium offerings or further data entry errors.

3.5 Recommendations

- Request that the data owner add a discovery/outreach channel field to future exports; without it, any question about acquisition or marketing effectiveness cannot be answered from this dataset.
- Prioritize completing the location field for the 387 unspecified records, since location is currently one of the more usable fields in the dataset and this gap limits confidence in geographic analysis.
- Carry the fee_flag and quality_review_flag columns from Week 1 into any Week 3 dashboard as filterable fields, so reviewers can toggle flagged records in or out rather than having them silently included.
- Consider whether Internship's scholarship value should be revisited given its dominant share of volume, if scholarship allocation is intended to reflect program priority.
- Correct the residual placeholder value found in the location field ("Quis voluptate totam") as a small follow-up to the Week 1 cleaning pass.
- Confirm with the data owner whether microscholarship carries its own currency/unit, and add a dedicated field for it if so this should be resolved before scholarship comparisons in this report are used for any funding or budget decision.

4. Conclusion

This exploratory analysis surfaced clear patterns in when opportunities are added, where they are hosted, and how scholarship value is distributed across categories; while also surfacing two genuine data gaps (missing location values and the absence of an outreach channel field) that should be addressed before deeper reporting or dashboarding work begins. Team 1 recommends these findings, along with the location and outlier data-quality notes, be carried forward into the Week 3 visualization and dashboard stage.

Appendix A: Team Charter

Team 1's full working charter, covering team roles, meeting format, tools used, and Week 1/Week 2 contribution breakdowns, is maintained as a separate document rather than reproduced here, in order to keep team members' contact details out of this shared report.

View the Team 1 Charter: [Team1 Team Charter.pdf](#)